SMA-ACC Network Guides Flexible Strategy Selection in Rats

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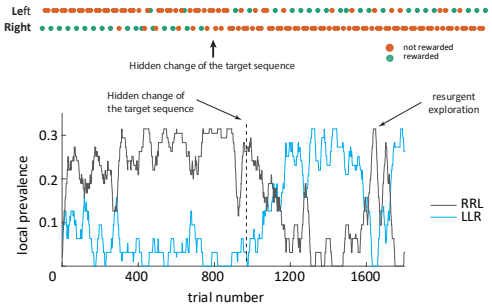
Abstract

The ability to efficiently acquire and organize knowledge is one of core aspect of general intelligence. It often requires animals to make inferences and test specific hypotheses about the latent structured relationships in the environment. To study how animals may approach the problem and how the brain may represent the inferred structure, we designed an experimental framework that requires animals to discover a latent sequence of actions that results in reward, and do it through self-guided exploration of sequences generated by a set of binary actions. We use pharmacological inactivation to demonstrate functional involvement in this task of the rodent homologues of Anterior Cingulate Cortex (ACC) and Supplementary Motor Area (SMA). We found that ACC ensemble representation associated with specific behavioral strategy is modulated depending on the context: whether it is a dominant target sequence or it is part of an exploratory bout. In contrast, SMA ensemble activity associated with strategy repre-sentation remains largely stable across the contexts. Targeted optogenetic suppression of SMA-ACC projection at strategy re-evaluation point increased the probability of the target sequence being repeated, but only in cases when animals encountered an unrewarded instance of that sequence. Our findings suggest that SMA-ACC network may be arranged in a way where SMA provides a stable basis of sequence representation that ACC utilizes at the moments of uncertainty and the internally-guided need for further exploration.

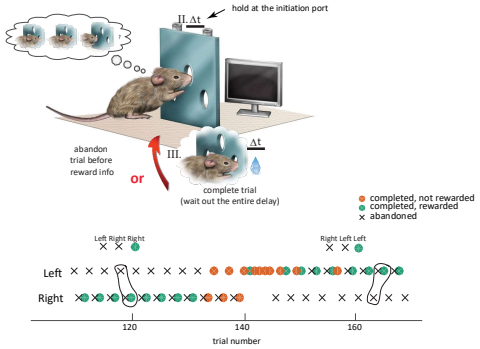
Task that exposes strategic action sequencing

Rats self-initiate trials, and report each choice by activating left (L) or right (R) side port. The animal has to discover that specific latent pattern (e.g. LLR) is preferentially rewarded.

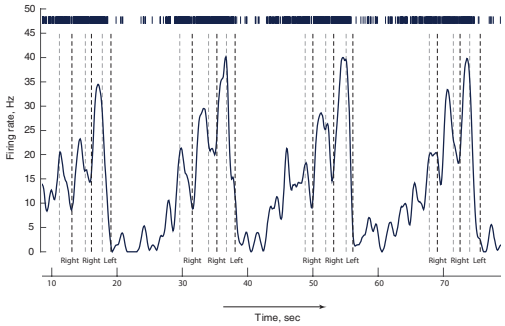


Animals adapt to unsignalled changes in the identity of the target sequence. Note also, however, that even when the target sequence is locally rewarded 100% of the time, unprompted bouts of previously rewarded sequence resergence is often observed when the enviornment is dynamic.

Rats indeed strategically sequence their actions

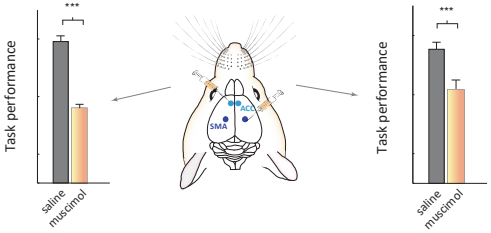


Rationale: get animals to pause specifically at the end of the action sequence they want to evaluate. Solution: task modification that introduces a long (3-10 sec) delay between side port entry and trial outcome availability, but also permits animals to abandon the wait and proceed with initiating the next trial. Note preferential pausing at the end of the target sequence within period of stability, and, transiently, on evry trial, once a chance is detected.

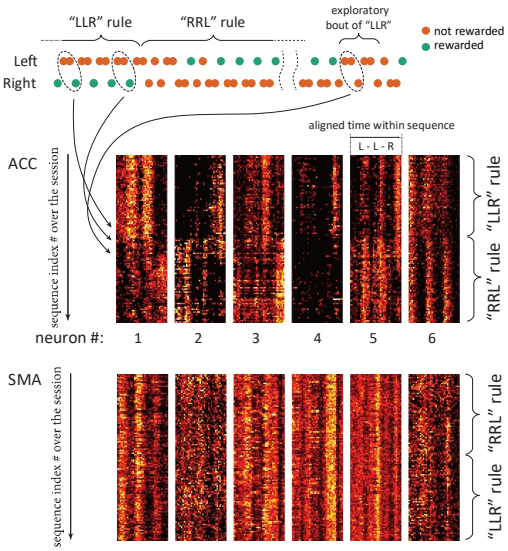


Concomitant emergence of pronounced ramping activity in ACC neurons with strong sequence selectivity as animals discover the target sequence is consistent with the behavioral chunking of choices into distinct sequences.

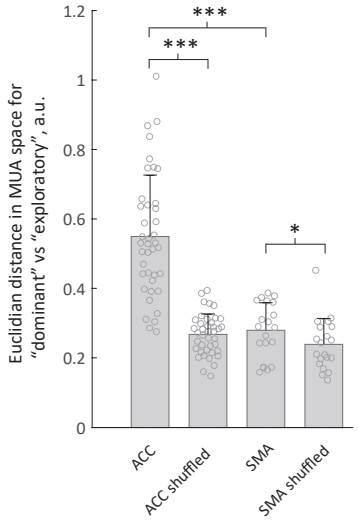
Necessity of ACC and SMA for the task performance



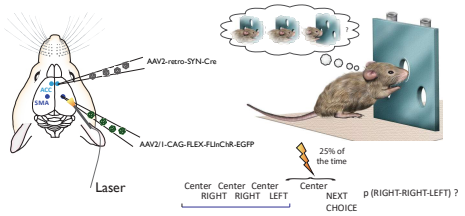
Context-dependent modulation of enesemble activity in ACC but not SMA



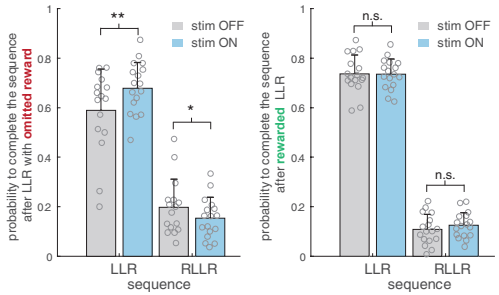
The sequence representation in ACC is modulated depending on the context: whether it is a dominant target sequence or it is part of an exploratory bout. In contrast, SMA ensemble activity associated with strategy representation remains largely stable across the contexts.



Silencing SMA-ACC projection at the time of strategy selection affects task performance



Suppression of SMA to ACC projection biases the animal to repeat the target sequence, but only following omitted reward when the animal may need to consider alternative options.



Conclusions

In the context of a behavioral framework that encourages animals to discovered sequential structure through self-guided exploration of individual sequences of actions, we uncovered strong sequence-related responses in ACC, and exposed one re-vealing organizational principle of ACC network dynamics: a marked modulation of neural representation for individual sequences based on their prevalence. The most striking modulation is based on whether a sequence represents the dominant strategy, or is evaluated in a transient burst of resurgent exploration, with an additional modulation by the local prevalence of that sequence within any behavioral epoch. Importantly, such marked shifts in encoding do not appear to be a brain-wide phenomenon. Indeed, even in a presumed homologue of the supplementary motor area (SMA), activity associated with a sequence remains stable across contexts.

The stability of sequence representation in SMA suggests an intriguing possibility that SMA provides a stable basis for sequence representation that ACC utilizes in different ways depending on the inferred behavioral context and the internally-guided need for further exploration.

Aknowledgements

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